UNITED UNIVERSITY

MINI PROJECT REPORT

ON

TOPIC:REAL STATE MARKET ANALYSIS AND PRICE PREDICTION

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Real Estate Market Analysis and Price Prediction

Introduction

The real estate market changes based on location, demand, economic conditions, and property features. Buyers and investors often find it difficult to estimate property prices accurately using manual methods. This project uses data analysis and machine learning to study housing market trends and predict house prices more efficiently.

Summary of the project

This project analyzes housing market trends using data analysis and machine learning. The dataset includes features like location, bedrooms, bathrooms, and square footage. Python libraries such as Pandas, NumPy, and Scikit-Learn are used in the project. Visualizations help understand property price patterns and market behavior. The Random Forest model predicts house prices with high accuracy. The project supports better real estate investment and pricing decisions.

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Objective

The objective of this project is to analyze housing market data and identify the factors affecting property prices. The project also aims to create visualizations and build a machine learning model for accurate price prediction

Tools & Technologies

- Python
- Pandas
- NumPy
- Matplotlib
- Plotly Express

and Scikit-Learn are used in this project. Jupyter Notebook or Google Colab is used as the development environment.

Dataset Description

The dataset is collected from Kaggle housing datasets. Important features include square footage, number of bedrooms and bathrooms, neighborhood, and year built. The target variable is Sale Price.

Methodology

The project involves data cleaning, feature engineering, data splitting, model training, and evaluation. The Random Forest Regressor is used to predict housing prices, and model performance is measured using MAE and R^2 score.

Visualization

Heatmaps, scatter plots, box plots, and histograms are used to understand housing market trends and identify relationships between features and property prices.

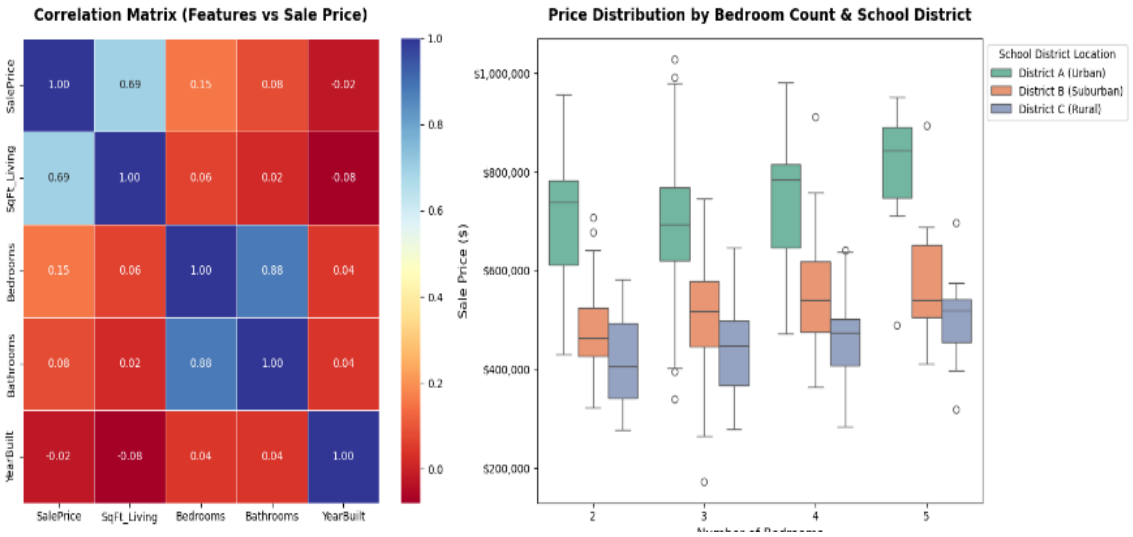
- **Heatmaps**: Show correlation matrix between house features and the final sale price.

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# 1. Generate realistic synthetic data for demonstration
np.random.seed(42)
n_houses = 400
sqft = np.random.normal(2100, 500, n_houses).astype(int)
bedrooms = np.random.choice([2, 3, 4, 5], size=n_houses, p=[0.15, 0.50, 0.25, 0.10])
bathrooms = (bedrooms * 0.55 + np.random.normal(0, 0.3, n_houses)).round(1)
bathrooms = np.clip(bathrooms, 1, 4.5)
year_built = np.random.randint(1960, 2025, n_houses)
districts = np.random.choice(['District A (Urban)', 'District B (Suburban)', 'District C (Rural)'], size=n_houses)
# SalePrice generation formula with added noise
base_price = 120000
price = (base_price + (sqft * 165) + (bedrooms * 21000) + ((2026 - year_built) * -450) + np.random.normal(0, 35000, n_houses))
price = np.where(districts == 'District A (Urban)', price * 1.35, price)
price = np.where(districts == 'District C (Rural)', price * 0.85, price)

df = pd.DataFrame({'SalePrice': price.astype(int),'SqFt_Living': sqft,'Bedrooms': bedrooms,'Bathrooms': bathrooms,'YearBuilt': year_built,
                   'School_District': districts})
# 2. Plotting the Visualizations Side-by-Side
fig, axes = plt.subplots(1, 2, figsize=(16, 6.5))
# Graph 1: Correlation Matrix Heatmap
corr = df[['SalePrice', 'SqFt_Living', 'Bedrooms', 'Bathrooms', 'YearBuilt']].corr()
sns.heatmap(corr, annot=True, cmap='RdYlBu', fmt=".2f", linewidths=0.7, cbar=True, ax=axes[0])
axes[0].set_title('Correlation Matrix (Features vs Sale Price)', fontsize=14, fontweight='bold', pad=15)

# Graph 2: Box Plot for Price Distributions
sns.boxplot(x='Bedrooms', y='SalePrice', hue='School_District', data=df, palette='Set2', ax=axes[1])
axes[1].set_title('Price Distribution by Bedroom Count & School District', fontsize=14, fontweight='bold', pad=15)
axes[1].set_xlabel('Number of Bedrooms', fontsize=12)
axes[1].set_ylabel('Sale Price ($)', fontsize=12)
axes[1].yaxis.set_major_formatter('${x:,.0f}')
axes[1].legend(title='School District Location', bbox_to_anchor=(1, 1), loc='upper left')
plt.tight_layout()
plt.show()
```



- **Geospatial Maps**: Plot interactive scatter maps showing price density across different zip codes.
- **Box Plots**: Compare price distributions across different bedroom counts or school districts

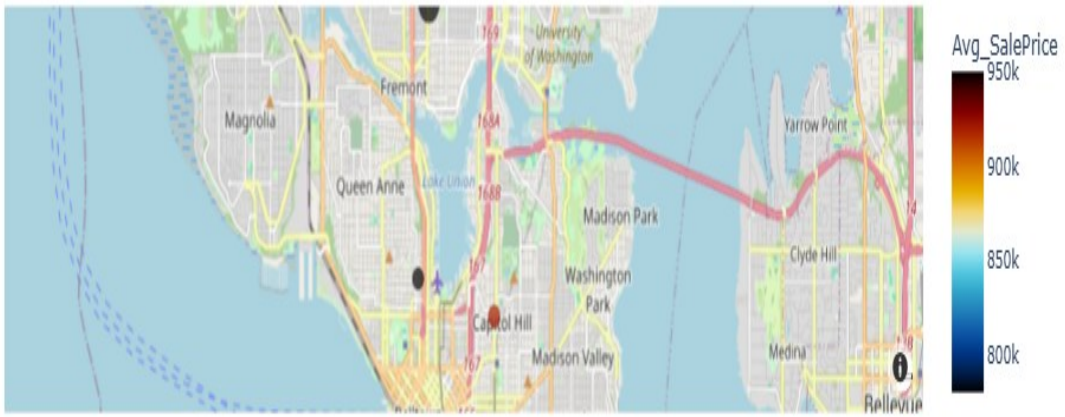
```python
import plotly.express as px

# Sample DataFrame containing regional latitude, longitude, and zip code metrics
geo_df = pd.DataFrame({
    'Zip_Code': ['98101', '98102', '98103', '98105', '98109'],
    'Latitude': [47.6085, 47.6245, 47.6601, 47.6614, 47.6288],
    'Longitude': [-122.3395, -122.3211, -122.3421, -122.3059, -122.3456],
    'Avg_SalePrice': [850000, 920000, 780000, 890000, 950000],
    'House_Count': [45, 32, 89, 54, 29]
})

# Generate interactive Mapbox scatter layer plot
fig = px.scatter_mapbox(geo_df,
                        lat="Latitude",
                        lon="Longitude",
                        color="Avg_SalePrice",
                        size="House_Count",
                        hover_name="Zip_Code",
                        color_continuous_scale=px.colors.cyclical.IceFire,
                        size_max=15,
                        zoom=11,
                        title="Interactive Real Estate Price Density by Zip Code")

fig.update_layout(mapbox_style="open-street-map")
fig.show()
```

Interactive Real Estate Price Density by Zip Code



- **Scatter Plot:** This graph directly shows how size changes value. It uses a colour gradient to track the age of the house and adds a trend line.

```python
import matplotlib.pyplot as plt
import seaborn as sns

# Set style theme
sns.set_theme(style="whitegrid")
plt.figure(figsize=(10, 6))

# Create scatter plot tracking size, price, and age
scatter = plt.scatter(df['SqFt_Living'], df['SalePrice'],
                      c=df['YearBuilt'], cmap='viridis',
                      alpha=0.7, edgecolors='w', s=70)

# Add a linear trend line
sns.regplot(data=df, x='SqFt_Living', y='SalePrice',
            scatter=False, color='red')

# Format titles and labels
plt.title('Impact of Living Space (SqFt) on Sale Price', fontsize=14, fontweight='bold', pad=15)
plt.xlabel('Interior Living Space (SqFt)', fontsize=12)
plt.ylabel('Sale Price ($)', fontsize=12)

# Format currency axes
plt.gca().yaxis.set_major_formatter('${x:,.0f}')

# Add color bar legend for house age
cbar = plt.colorbar(scatter)
cbar.set_label('Year House Was Built', fontsize=11)

plt.tight_layout()
plt.show()
```

(Histogram + KDE): This graph helps you see if your market dataset skews toward budget housing or luxury estates.

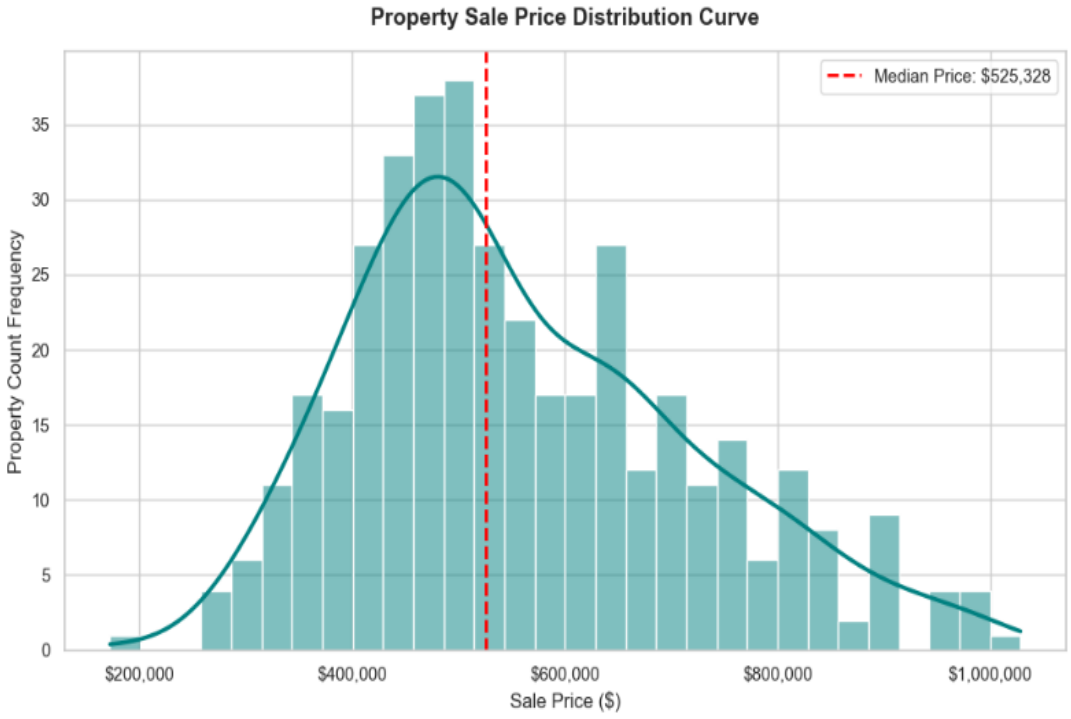
```
plt.figure(figsize=(10, 6))

# Plot histogram overlayed with a smooth Kernel Density Estimate curve
sns.histplot(df['SalePrice'], kde=True, color='teal', bins=30, line_kws={'linewidth': 2.5})

# Add a statistical vertical line for the median price marker
median_price = df['SalePrice'].median()
plt.axvline(median_price, color='red', linestyle='--', linewidth=2,
            label=f'Median Price: ${median_price:,.0f}')

# Label configurations
plt.title('Property Sale Price Distribution Curve', fontsize=14, fontweight='bold', pad=15)
plt.xlabel('Sale Price ($)', fontsize=12)
plt.ylabel('Property Count Frequency', fontsize=12)
plt.gca().xaxis.set_major_formatter('${x:,.0f}')
plt.legend(fontsize=11)

plt.tight_layout()
plt.show()
```



Results & Insights

The project found that square footage and location are the most important factors affecting house prices. The Random Forest model achieved high prediction accuracy with an R^2 score of around 0.88.

Conclusion

The project demonstrates how machine learning and data analysis improve real estate price prediction. Future improvements may include adding real-time economic data and advanced prediction models.

REFERENCES

- **McKinney, W. (2017). Python for Data Analysis Data Wrangling with Pandas, NumPy, and Python.**

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